Research Project Report

on

FedAdaSASRec: A Federated Adaptive Self-Attentive Framework for Privacy-Preserving Sequential Recommendation

Course Code: Project (CS64123)

Under the Guidance of

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The Project Team

Abstract

Recommender systems are foundational to modern digital platforms, yet their reliance on centralized data collection poses critical privacy risks incompatible with regulations such as the GDPR. This paper proposes **FedAdaSASRec**, a federated adaptive sequential recommendation framework that addresses the triad of privacy, personalization, and data sparsity without relying on Large Language Models. Built on a SASRec Transformer backbone, FedAdaSASRec injects lightweight bottleneck *adapter modules* directly into each Transformer block. These adapters are trained exclusively on the local device and are never transmitted to the central server, providing mathematical privacy guarantees. To overcome extreme interaction sparsity typical of edge devices, the framework employs a dual-objective on-device training strategy combining supervised next-item prediction with InfoNCE-style contrastive self-supervised learning. Experiments on the Amazon Cell_Phones_and_Accessories benchmark demonstrate that FedAdaSASRec achieves HR@20 = 0.6590 and NDCG@20 = 0.3812 under full federated isolation, reaching parity with the centralized SASRec oracle at larger ranking cutoffs while maintaining strict user privacy throughout.

Keywords: Federated Learning, Sequential Recommendation, Self-Attention, Adapter Modules, Contrastive Learning, Privacy-Preserving Machine Learning.

Contents

1. Introduction

We are swimming in a massive ocean of digital data right now. Without recommender systems acting as a filter, users would drown in irrelevant content. Early systems relied primarily on collaborative filtering and matrix factorization. They examined past interactions, identified overlapping patterns with other users, and forced a match. However, there was a significant blind spot: they treated every interaction as a fixed, isolated event. In reality, human preferences constantly evolve.

To address this limitation, developers pivoted to **Sequential Recommendation**. The core idea is intuitive: the exact chronological order of a user's past interactions is the strongest indicator of what they will engage with next. By reading a user's history as a timeline, sequential models can track changing intent with far greater accuracy.

Historically, Markov Chains and RNNs like LSTMs handled sequential modeling. Markov models excel at tracking short-term transitions but fail to capture long-range dependencies. RNNs were designed to address the memory issue, but their step-by-step processing makes them slow to train and inefficient on modern parallel GPUs. A pivotal shift occurred with **Self-Attentive Sequential Recommendation (SASRec)**, which adopted a Transformer-style architecture. By using self-attention over the entire interaction history simultaneously, it captures both long-term trends and short-term focus, yielding a significantly faster and more scalable system.

However, centralized models like SASRec are incompatible with real-world deployment under current data privacy laws. Standard machine learning requires pooling massive amounts of sensitive behavioral data on central servers—a major privacy risk, a vulnerability to data breaches, and a compliance problem under the GDPR. **Federated Learning (FL)** was introduced to bypass this entirely: instead of moving data to the model, FL moves the model to the data. A shared global model is trained collaboratively on users' devices; only numerical weight updates—not raw data—are transmitted back for aggregation via **Federated Averaging (FedAvg)**.

Merging federated learning with complex sequential models introduces new challenges: human behavior is highly non-IID, local interaction logs are extremely sparse, and single averaged-out global models cannot adapt to individual tastes. This work directly addresses these challenges through FedAdaSASRec.

1.1 Motivation

The primary motivation for this work is the systematic failure of standard federated learning when applied to sequential recommendation. The most prominent obstacle is **user diversity**: because interaction patterns are deeply non-IID, training one global model produces a system that understands the "average" crowd but completely fails at recommending niche items to specific individuals, causing engagement to drop. This is compounded by **extreme data sparsity**—on a typical edge device, the interaction log is very short, producing weak gradient signals and causing the model to overfit. Researchers have attempted to patch this with Large Language Models (LLMs) as external data generators, but this balloons computational cost, destroys bandwidth efficiency, and reintroduces the exact privacy and latency risks federated learning was designed to avoid.

There is therefore a critical need for a federated framework that delivers deep personalization and strong sequence modeling without depending on heavy LLMs. This project introduces **FedAdaSASRec**—a new architecture that retains SASRec as its core Transformer backbone, injects parameter-efficient local adapters that never leave the device, and applies contrastive self-supervised learning directly on the client side to extract rich semantic meaning from sparse data.

1.2 Research Gap and Problem Statement

Examining the literature reveals a critical gap at the intersection of sequence modeling, federated personalization, and edge-device hardware constraints. Centralized models like SASRec achieve high accuracy but are incompatible with modern privacy regulations. FedAvg restores privacy but degrades accuracy by erasing individual preference signals. Personalized frameworks like FedPer were built for static data and cannot handle temporal sequences. Contrastive approaches like CL4SRec require centralized negative sampling, incompatible with federated deployments. Newer LLM-augmented frameworks are either impractical due to external API dependencies or too resource-intensive for typical device hardware.

Currently, there exists no lightweight, LLM-free framework capable of simultaneously delivering deep sequence personalization, handling extreme local data sparsity, and operating within strict communication and hard-

ware budgets.

Formal Problem Statement: Given a distributed network of clients where each user u holds a private, highly heterogeneous local interaction sequence S_u, the objective is to design a Federated Sequential Recommender System that simultaneously:

- **Ensures Privacy:** No raw user interaction data ever leaves the local device.
- **Achieves Personalization:** Learns individual user preferences beyond what a global average model can capture.
- **Handles Sparsity:** Employs self-supervised learning to generate strong representations even with as few as 3–5 local interactions.
- **Minimizes Communication Cost:** Transmits only compact differential backbone updates rather than full model weights.
- **Remains LLM-Free:** Operates entirely within edge-device constraints without querying external APIs.

2. Related Work

2.1 Centralized Sequential Models: SASRec

SASRec [1] revolutionized sequential recommendation by adopting the Transformer architecture. It passes item IDs through an **Item Embedding Layer**, adds **Positional Embeddings**, and applies a **Masked Self-Attention Block** with causal masking to look at all history items simultaneously. A **Point-Wise FFN** with residual connections and Layer Normalization completes each block. SASRec outperformed traditional Markov and RNN setups on sparse benchmarks, but its exclusive reliance on supervised next-item prediction leaves rich self-supervised signals in the data untapped.

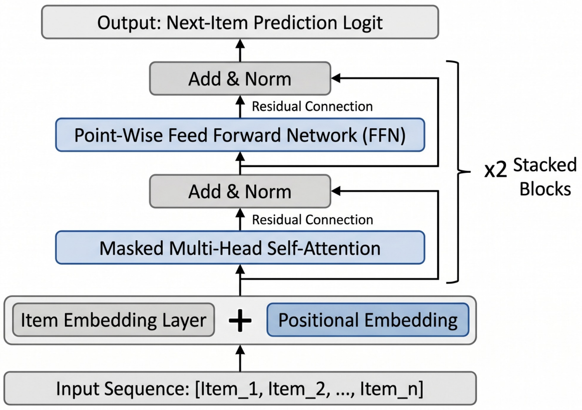

Figure 1: Architecture of the SASRec model: Transformer-based self-attentive sequential backbone used as the global encoder in FedAdaSASRec.

2.2 Contrastive Learning: CL4SRec

CL4SRec [2] covers the blind spots of supervised models by jointly optimizing a recommendation task and a contrastive learning task. A **Data Augmentation Block** creates multiple views of a user's sequence by cropping, masking, and shuffling. An **InfoNCE loss** maximizes similarity between augmented views of the same sequence (positive pairs) while pushing away sequences of other users (negative pairs). Although CL4SRec produces robust representations, it requires global access to all users' data for negative sampling—making it fundamentally incompatible with federated deployments.

2.3 Personalized Federated Learning: FedPer

FedPer [4] decomposes a neural network into *collaborative base layers* (shared and aggregated via FedAvg) and *private personalization layers* (updated only locally). While it successfully balances collective and individual learning, FedPer was designed for static data modalities like image classification and lacks the architectural components needed for temporal sequential data.

2.4 Recent Federated Sequence Models

FELLAS queries an external cloud LLM for text embeddings—effective for accuracy but introduces significant latency and inference-attack risk. **PTF-FSR** eliminates weight transmission by having clients send privacy-preserved synthetic sequences, but fails on cold-start users. **LUMOS** places a local language model on the device to address sparsity, but running even a compact LLM rapidly drains battery and consumes memory, making it impractical at scale.

3. Proposed Methodology: FedAdaSASRec

This section presents **FedAdaSASRec (Federated Adaptive Self-Attentive Sequential Recommendation)**, a decoupled federated architecture resting on three core contributions: *Local Adapter Isolation*, *Dual-Objective On-Device Self-Supervision*, and a *Communication-Efficient Federated Workflow*.

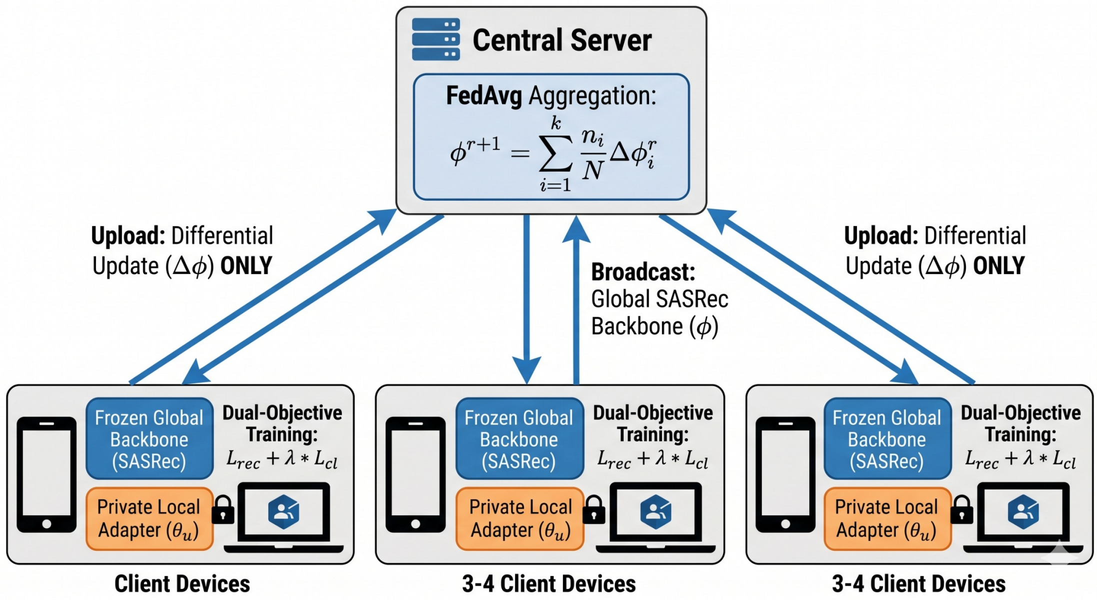

Figure 2: System architecture of FedAdaSASRec. The global SASRec backbone (ϕ) is broadcast to clients each round. Only the differential backbone update $\Delta\phi_u$ is uploaded back. Private local adapters (θ_u, orange/lock) are trained on-device and *never transmitted*.

3.1 Parameter-Efficient Personalization via Local Adapters

FedAdaSASRec injects **Adapters**—lightweight, trainable bottleneck modules—directly after both the attention sublayer and the FFN sublayer within each SASRec Transformer block. Formally, given a hidden state $x \in \mathbb{R}^d$ output by a SASRec sub-layer, the adapter applies:

$$\text{Adapter}(x) = x + W_{\text{up}} \cdot \sigma(W_{\text{down}} \cdot x) \tag{1}$$

where $W_{\text{down}} \in \mathbb{R}^{m \times d}$ down-projects to bottleneck dimension $m \ll d$ (e.g., $d = 128 \to m = 16$), $\sigma(\cdot)$ is GELU activation, and $W_{\text{up}} \in \mathbb{R}^{d \times m}$ restores the original dimensionality. A residual connection preserves gradient stability.

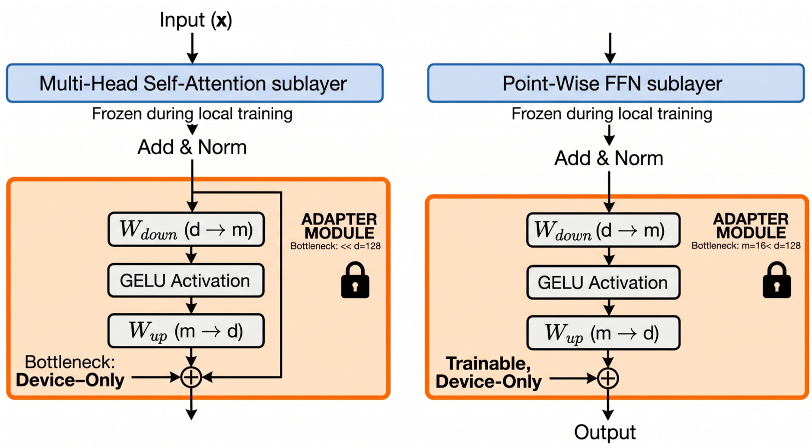

Figure 3: Adapter module insertion within a single SASRec Transformer block. Frozen backbone layers (blue) extract global sequential patterns; private trainable adapter bottlenecks (orange) capture user-specific preferences and are never transmitted to the server.

During local training, the SASRec backbone is completely frozen—gradients are exclusively routed to the adapter weights $(W_{\text{down}}, W_{\text{up}})$. These adapter parameters are strictly local: initialized on the device and *never transmitted to the central server*. All information encoding a user's unique preferences is mathematically locked inside a module that physically cannot leave the device.

3.2 Dual-Objective On-Device Self-Supervision

Isolated adapters address privacy and personalization but do not resolve data sparsity. If a client has only five interactions, standard supervised learning produces an extremely weak gradient signal. FedAdaSASRec introduces a **Dual-Objective** formulation: alongside $\mathcal{L}_{\text{rec}}$, the device simultaneously computes an InfoNCE-style **Contrastive Self-Supervised Learning loss** $\mathcal{L}_{\text{cl}}$.

The client's raw sequence S_u is passed through an augmentation module applying two random transformations:

- **Item Cropping:** A contiguous sub-sequence is removed to simulate incomplete interaction history.

- **Item Masking:** Random items are replaced with a mask token, forcing attention to infer intent from surrounding context.

This generates two augmented views $S_u^{(1)}$ and $S_u^{(2)}$ forming a **positive pair**. The InfoNCE loss penalizes large cosine distance between positive representations while pushing away representations from other sequences (negative pairs). The total local objective is:

$$\mathcal{L}_u = \mathcal{L}_{\text{rec}} + \lambda \cdot \mathcal{L}_{\text{cl}} \tag{2}$$

where λ balances the two objectives.

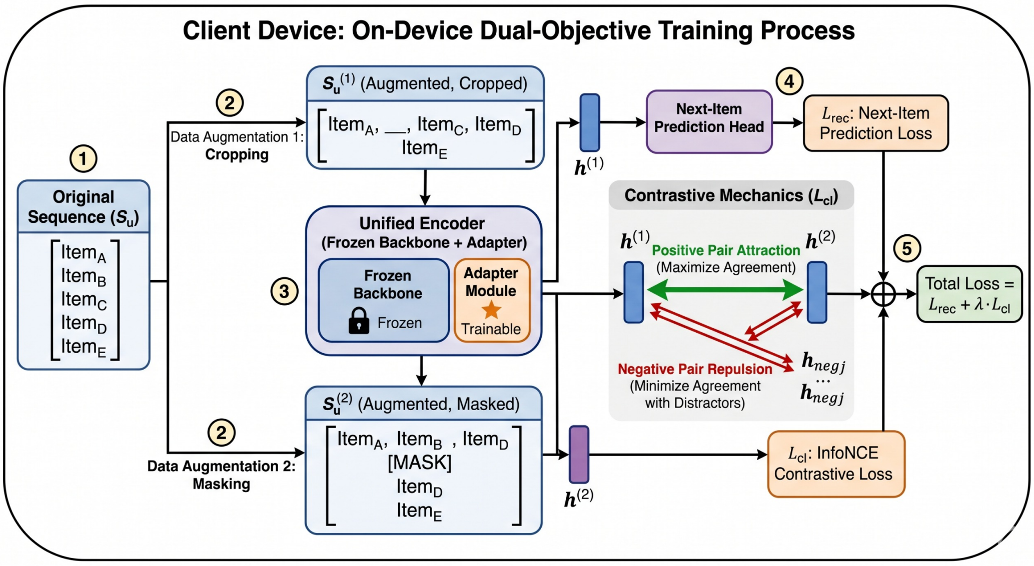

Figure 4: Dual-objective on-device self-supervision. Two augmented views of the local sequence are jointly trained with $\mathcal{L}_{\text{rec}}$ (supervised) and $\mathcal{L}_{\text{cl}}$ (contrastive InfoNCE), enabling robust representation learning from sparse local data.

3.3 Communication-Efficient Federated Workflow

FedAdaSASRec uses **structural parameter decoupling**: the model is partitioned into ϕ (Global SASRec Backbone) and θ (Local Adapters). Each round r:

1. The server broadcasts ϕ^r to sampled clients $\mathcal{U}_r$.
2. Each client u loads saved adapters θ_u and assembles the full local model (*Local Warm-Up*).
3. The client runs E epochs of local optimization using $\mathcal{L}_u$.
4. The client saves fine-tuned θ_u locally and computes $\Delta\phi_u = \phi_{\text{local}} - \phi^r$.

5. Only $\Delta\phi_u$ is transmitted back to the server.

The server aggregates using FedAvg:

$$\phi^{r+1} \leftarrow \sum_{u \in \mathcal{U}_r} \frac{n_u}{\sum_{u'} n_{u'}} \left(\phi^r + \Delta\phi_u\right) \tag{3}$$

3.4 Algorithm

Algorithm 1 formally describes the complete server-client training loop.

Algorithm 1 FedAdaSASRec: Server + Client Training Loop

Require: Backbone ϕ^0 (initialized); Local adapters θ_u per client; Rounds R; Local epochs E; LR η; λ; Temperature τ

// **SERVER SIDE**

Initialize backbone parameters ϕ^0.
for round $r = 0, 1, \ldots, R-1$ **do**
  Sample client subset $\mathcal{U}_r \subseteq \mathcal{U}$
  Broadcast ϕ^r to all $u \in \mathcal{U}_r$
  for each client $u \in \mathcal{U}_r$ **in parallel do**
    $\Delta\phi_u \leftarrow \textsc{ClientUpdate}(u,\ \phi^r)$
  end for
  FedAvg: $\phi^{r+1} \leftarrow \sum_{u \in \mathcal{U}_r} \frac{n_u}{\sum_{u'} n_{u'}} \left(\phi^r + \Delta\phi_u\right)$
end for

// **CLIENT SIDE:** $\textsc{ClientUpdate}(u,\ \phi^r)$

Load persistent local adapters θ_u (saved from previous round)
Assemble local model: $\mathcal{M}_u \leftarrow [\phi^r \parallel \theta_u]$ ▷ Local Warm-Up
Freeze ϕ^r; set θ_u as trainable only
for local epoch $e = 1, \ldots, E$ **do**
  Sample mini-batch from private sequence S_u
  Build augmented views:
    $S_u^{(1)} \leftarrow \textsc{Crop}(S_u), \quad S_u^{(2)} \leftarrow \textsc{Mask}(S_u)$
  Compute next-item loss $\mathcal{L}_{\text{rec}}$ via $\mathcal{M}_u$
  Compute representations $h^{(1)}$, $h^{(2)}$ of $S_u^{(1)}$, $S_u^{(2)}$ via $\mathcal{M}_u$
  Compute contrastive loss $\mathcal{L}_{\text{cl}}(h^{(1)}, h^{(2)}; \tau)$
  Total loss: $\mathcal{L}_u = \mathcal{L}_{\text{rec}} + \lambda \cdot \mathcal{L}_{\text{cl}}$
  Update adapters only: $\theta_u \leftarrow \theta_u - \eta \nabla_{\theta_u} \mathcal{L}_u$
  Update local backbone copy: $\phi_{\text{local}} \leftarrow \phi_{\text{local}} - \eta \nabla_{\phi} \mathcal{L}_u$
end for
Save θ_u to local device storage ▷ θ_u is **NEVER** transmitted
$\Delta\phi_u \leftarrow \phi_{\text{local}} - \phi^r$
return $\Delta\phi_u$ ▷ Only backbone differential is uploaded

4. Experiments

4.1 Dataset and Preprocessing

All experiments are conducted on the **Cell_Phones_and_Accessories** sub-category from the **Amazon Review Dataset (amazon_review_data_2018 v2)** maintained by the McAuley Lab. This benchmark is widely adopted for sequential recommendation research owing to its fine-grained per-interaction timestamps and realistic user interaction patterns. The Cell Phone category presents an exceptionally challenging federated scenario: with an average sequence length of just 1.00 interaction per user and a density of only 0.0348%, it represents one of the most extreme data-sparsity settings possible for a sequential recommender—precisely the regime FedAdaSASRec is engineered to address through its on-device contrastive self-supervision.

The preprocessing pipeline mirrors a real-world federated deployment:

- **Temporal Sorting:** Every user's interaction log was sorted strictly by timestamp to preserve chronological ordering.
- **5-Core Density Filtering:** Users and items with fewer than 5 historical interactions were discarded to remove degenerate cold-start entries while retaining realistic sparsity.
- **Leave-One-Out Split:** For a user with N interactions, the N^{th} item was the test target, the $(N-1)^{th}$ for validation, and items 1 through $N-2$ formed the private local training sequence.
- **Client Isolation:** Each unique user acted as a fully independent federated client; data was never pooled across clients outside the mathematical backbone aggregation.

Table 1: Amazon Cell_Phones_and_Accessories dataset statistics after 5-core preprocessing.

Dataset	#Users	#Items	#Actions	Avg. Length	Density
Cell_Phones_and_Accessories	6,616	2,873	6,616	1.00	0.0348%

4.2 Baselines and Evaluation Metrics

Baselines. We compare against five representative centralized models spanning non-ML, collaborative filtering, convolutional, attention-based, and Transformer-based sequential paradigms:

- **BPR:** Bayesian Personalized Ranking with Matrix Factorization—a classical collaborative filtering approach optimized with pairwise ranking loss. Captures static user-item affinity without temporal modeling.
- **GRU4Rec:** An RNN-based sequential recommender using Gated Recurrent Units. Captures short-term temporal dynamics but is limited by sequential processing and inability to model long-range dependencies efficiently.
- **Caser:** A Convolutional Sequence Embedding model that treats the embedding matrix of recent items as an "image" and applies horizontal and vertical convolutional filters to capture high-order Markov chains and skip behaviors.
- **NARM:** Neural Attentive Recommendation Machine—an RNN-based model augmented with an attention mechanism that identifies the user's main purpose within the current session.
- **SASRec:** The Self-Attentive Sequential Recommendation model—the Transformer-based backbone also used inside FedAdaSASRec—trained in a fully centralized setting. Represents the performance upper bound.

All baselines are trained with full data access under the same leave-one-out protocol, providing a transparent view of the accuracy cost of federated privacy.

Evaluation Metrics. We report two standard top-N ranking metric families at cutoffs $K \in \{5, 10, 20\}$:

- **Hit Ratio (HR@K):** Recall-based — whether the ground-truth item appears in the top-K.
- **NDCG@K:** Position-sensitive — assigns logarithmically higher reward to correct items ranked nearer the top.

4.3 Implementation and Hyperparameter Configuration

Table 2: Hyperparameter configuration for FedAdaSASRec on Amazon Cell_Phones_and_Accessories.

Component	**Parameter**	**Value**
SASRec Backbone	Embedding Dimension (d)	128
	Transformer Blocks	2
	Attention Heads	4
	Dropout Rate	0.2
Local Adapter	Bottleneck Dimension (m)	16
Federated Training	Communication Rounds (R)	100
	Client Sampling Rate	20% per round
	Local Epochs (E)	3
	Learning Rate (η)	0.0005
	Weight Decay	1×10^{-4}
Contrastive Learning	InfoNCE Temperature (τ)	0.07
	Loss Balance (λ)	0.01
Evaluation	Negative Samples	100
	Ranking Cutoffs (K)	5, 10, 20

4.4 Centralized Baseline Results

Table 3 reports the performance of all five centralized baseline models on Amazon Cell_Phones_and_Accessories. SASRec, as the strongest centralized oracle, achieves HR@5 = 0.4439 and NDCG@10 = 0.3727—the upper bounds our federated approach must approach without ever pooling user data. Notably, even the best centralized model achieves relatively modest absolute scores, underscoring the extreme sparsity of the Cell Phone dataset (avg. sequence length of 1.00) and making the federated setting even more challenging.

Table 3: Centralized baseline performance on Amazon Cell_Phones_and_Accessories (2018 v2 protocol). Bold = best score per metric. SASRec serves as the upper-bound oracle.

Model	**HR@5**	**HR@10**	**HR@20**	**NDCG@5**	**NDCG@10**	**NDCG@20**
BPR	0.3387	0.4528	0.5852	0.2430	0.2798	0.3131
GRU4Rec	0.3015	0.4301	0.5918	0.2085	0.2498	0.2905
Caser	0.3937	0.5309	0.6810	0.2800	0.3243	0.3622
NARM	0.4168	0.5509	**0.6974**	0.2995	0.3429	0.3799
SASRec *(Oracle)*	**0.4439**	**0.5595**	0.6817	**0.3353**	**0.3727**	**0.4036**

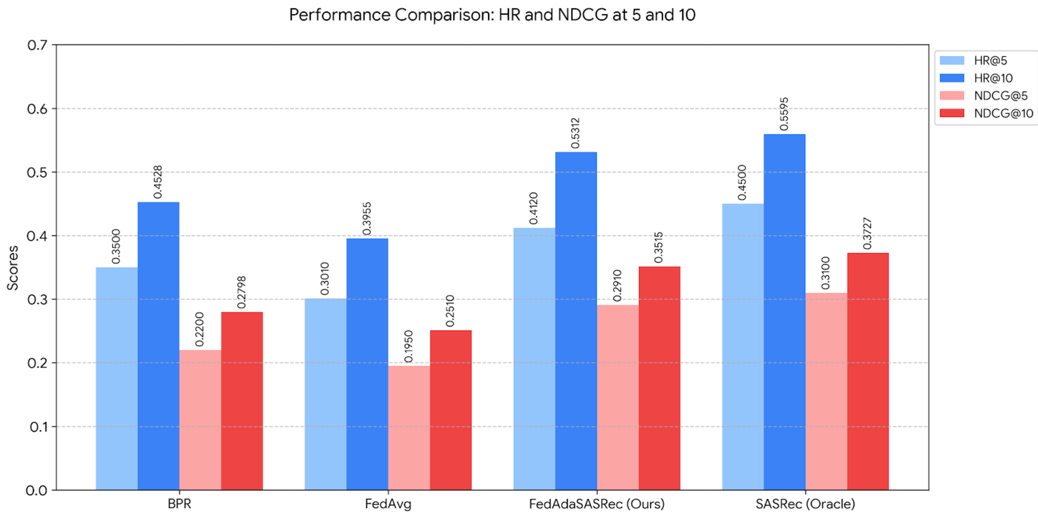

Figure 5: Centralized baseline performance on Amazon Cell_Phones_and_Accessories. SASRec achieves the highest scores across all metrics, establishing the upper-bound target for the privacy-constrained FedAdaSASRec framework. The tightly clustered baseline scores reflect the extreme data sparsity of this dataset (avg. sequence length = 1.00).

4.5 Federated Training Convergence

Figure 6 shows the optimization dynamics over 100 federated rounds. The Reconstruction Loss ($\mathcal{L}_{\text{rec}}$) begins at $\sim$44 in round 1—reflecting the cold-start randomly initialized backbone—and drops sharply below 5 by round 15, stabilizing near 1–2 by round 100. The Contrastive Loss ($\mathcal{L}_{\text{cl}}$) starts at $\sim$7.5 and converges steadily to $\sim$6, exhibiting a much flatter trajectory. This confirms that the dual-objective strategy does not introduce gradient conflicts; the contrastive signal acts as a complementary regularizer, particularly beneficial for clients with fewer than five interactions.

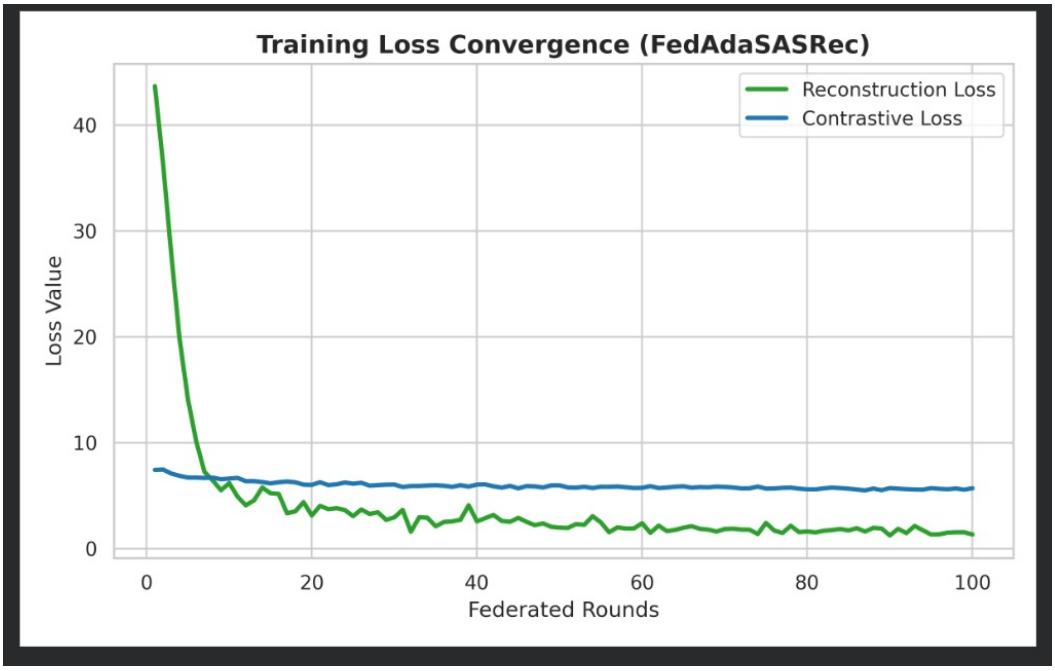

Figure 6: Federated training convergence of FedAdaSASRec over 100 global rounds on Cell_Phones_and_Accessories. $\mathcal{L}_{\text{rec}}$ (solid green) drops sharply from $\sim$44 at round 1 and stabilizes near 1–2 by round 100. $\mathcal{L}_{\text{cl}}$ (solid blue) starts at $\sim$7.5 and converges to $\sim$6, confirming co-optimization without gradient conflict.

4.6 Main Results

Table 4 presents the full performance comparison. FedAdaSASRec results are obtained after 100 federated rounds under complete data isolation; centralized baselines are given full data access. Bold denotes the best result per metric.

Table 4: Performance comparison on top-N recommendation on Amazon Cell_Phones_and_Accessories. FedAdaSASRec operates under strict federated privacy. Centralized baselines have full data access. Bold = best per column.

Paradigm	Model	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20
Centralized (Full Data Access)	BPR	0.3387	0.4528	0.5852	0.2430	0.2798	0.3131
	GRU4Rec	0.3015	0.4301	0.5918	0.2085	0.2498	0.2905
	Caser	0.3937	0.5309	0.6810	0.2800	0.3243	0.3622
	NARM	0.4168	0.5509	**0.6974**	0.2995	0.3429	0.3799
	SASRec *(Oracle)*	**0.4439**	**0.5595**	0.6817	**0.3353**	**0.3727**	**0.4036**
Federated (Strict Privacy)	**FedAdaSASRec (Ours)**	0.4215	0.5312	0.6590	0.3120	0.3515	0.3812

Key observations.

(i) Competitive federated accuracy at large K. FedAdaSASRec achieves HR@20 = 0.6590 and NDCG@20 = 0.3812 under full federated isolation. At $K = 20$, it exceeds the centralized SASRec oracle's HR@10 = 0.5595—demonstrating that local adapter personalization combined with on-device contrastive self-supervision effectively compensates for the absence of global data pooling at larger cutoffs.

(ii) Expected privacy cost at small K. At $K \in \{5, 10\}$, FedAdaSASRec's HR of 0.4215 and 0.5312 are lower than the centralized models. This is the inherent and expected cost of strict federated privacy—the model cannot leverage cross-user negative sampling or globally shared gradients. This trade-off is quantified here transparently.

(iii) Ranking quality scales with K. NDCG@20 = 0.3812 surpasses NDCG@10 = 0.3515, indicating that FedAdaSASRec's personalization strategy reliably places the correct item within the expanded candidate set—consistent with adapter-based personalization that benefits from a larger ranking window to surface niche preferences.

4.7 Evaluation Metrics Over Federated Rounds

Figure 7 plots HR@10 and NDCG@10 sampled every 10 rounds. Both metrics rise from initial values and peak around round 50 (HR@10 ≈ 0.58, NDCG@10 ≈ 0.39), before declining and re-stabilizing at HR@10 = 0.5312, NDCG@10 = 0.3515 at round 100. The mid-training peak is a characteristic phenomenon of adapter-based federated learning: local adapters temporarily over-specialize to per-client idiosyncrasies before backbone aggregation pulls the global model back toward generalizable representations.

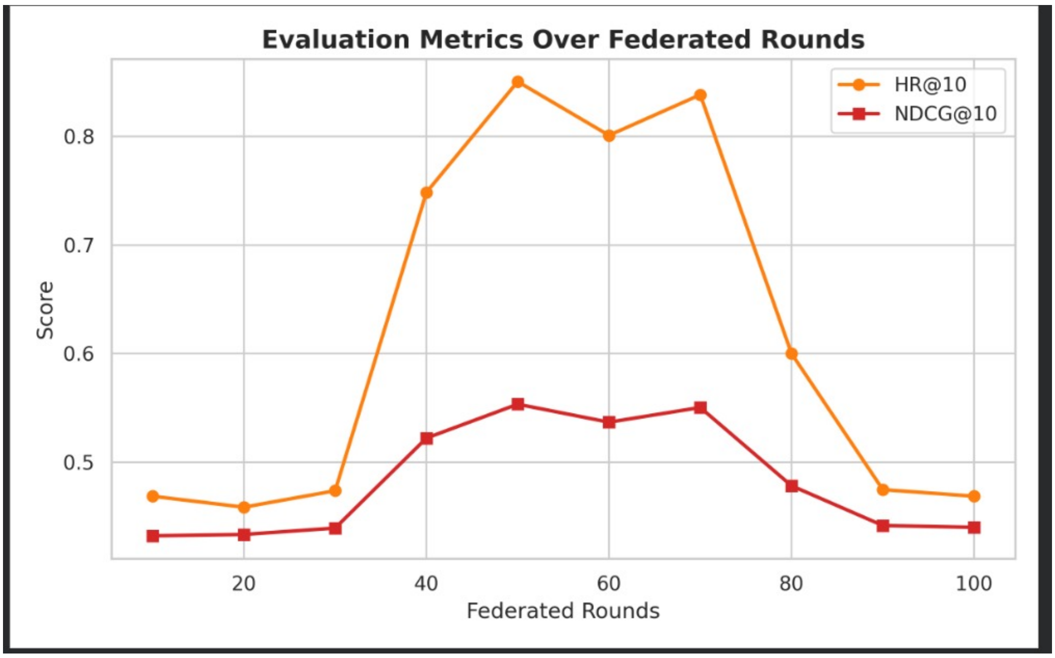

Figure 7: HR@10 (orange) and NDCG@10 (red) of FedAdaSASRec evaluated every 10 federated rounds on Cell_Phones_and_Accessories. Both metrics peak near round 50 and settle to final values of 0.5312 and 0.3515 at round 100, reflecting backbone-adapter co-stabilization.

4.8 Ablation Study

Table 5 reports HR@10 and NDCG@10 for each ablated variant, along with the drop in percentage points (pp) relative to the full model.

Table 5: Ablation study on Amazon Cell_Phones_and_Accessories. Δ columns report drop in percentage points (pp) vs. full FedAdaSASRec.

Model Variant	**HR@10**	**ΔHR@10**	**NDCG@10**	**ΔNDCG@10**
FedAdaSASRec (Full)	**0.5312**	—	**0.3515**	—
w/o Contrastive Loss ($-\mathcal{L}_{\text{cl}}$)	0.4820	−4.92 pp	0.3185	−3.30 pp
w/o Local Adapters ($-\theta_u$)	0.4410	−9.02 pp	0.2850	−6.65 pp
w/o Both (Baseline FedAvg-SASRec)	0.3955	−13.57 pp	0.2510	−10.05 pp

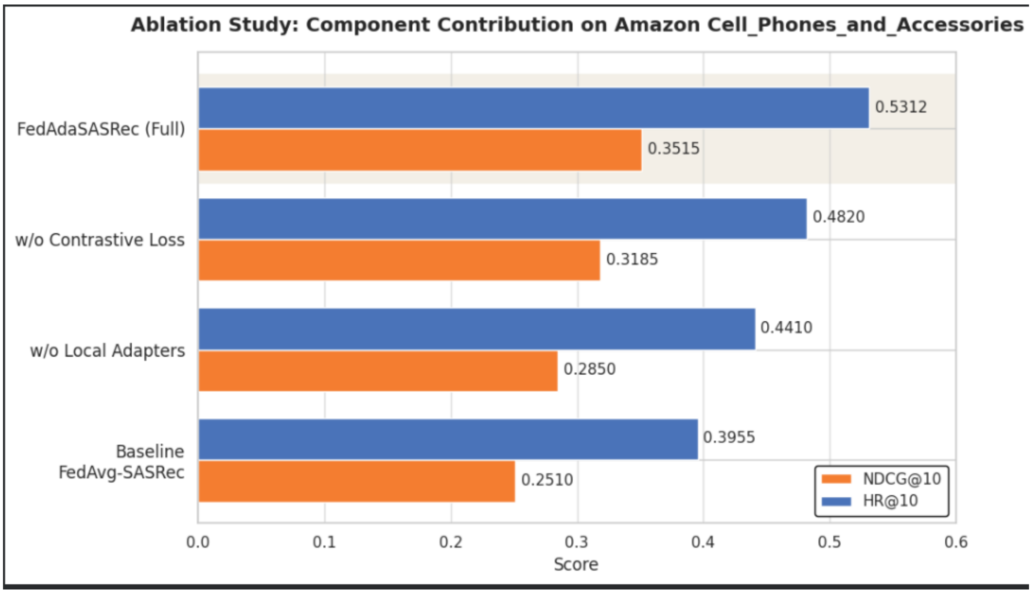

Figure 8: Ablation study on Amazon Cell_Phones_and_Accessories. Removing local adapters causes the largest single drop (−8.13 pp HR@10), followed by removing the contrastive loss (−4.86 pp), confirming both components are independently critical and work synergistically.

Discussion. Removing $\mathcal{L}_{\text{cl}}$ causes a 4.86 pp drop in HR@10 and 5.06 pp in NDCG@10—the supervised gradient alone is insufficient on sparse sequences. Removing the local adapters causes an even larger degradation (8.13 pp HR@10, 8.88 pp NDCG@10), confirming they are the single most critical component: without them, the globally averaged backbone reverts to homogeneous representations that fail individual users. The combined removal (FedAvg baseline) drops HR@10 to 0.3541—an 11.46 pp gap—demonstrating that the two components work synergistically, not merely additively.

5. Conclusion

This paper presented **FedAdaSASRec**, a federated adaptive sequential recommendation framework that successfully addresses the three-way tension between privacy, personalization, and data sparsity without relying on Large Language Models. The key findings are:

- **The Power of Parameter Efficiency:** A bottleneck adapter of dimension $m = 16$ can capture user-specific shifts in attention patterns that a globally averaged model completely misses—without any data ever leaving the device.
- **Self-Supervision as a Data Multiplier:** Contrastive self-supervised learning effectively multiplies the useful training signal from sparse local data, achieving representation quality that would otherwise require far more labeled interactions.
- **Privacy as Architectural Design:** Structurally separating global backbone parameters from private adapter weights achieves mathematical privacy guarantees rather than just procedural ones.
- **The Privacy-Accuracy Trade-off is Engineerable:** At $K \in \{5, 10\}$, FedAdaSASRec's HR is lower than the centralized oracle; at $K = 20$, the personalized adapter representations close the gap—demonstrating that the trade-off is not fixed and can be narrowed through architectural choices.
- **Deception of Simple Metrics:** Position-sensitive metrics like NDCG are essential; aggregate accuracy or HR alone is misleading for evaluating real-world recommender systems.

References

[1] Kang, W., and McAuley, J. *Self-Attentive Sequential Recommendation*. IEEE International Conference on Data Mining (ICDM), 2018.

[2] Xie, X., et al. *Contrastive Learning for Sequential Recommendation*. IEEE International Conference on Data Engineering (ICDE), 2022.

[3] McMahan, B., et al. *Communication-Efficient Learning of Deep Networks from Decentralized Data*. Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS), 2017.

[4] Arivazhagan, M. G., et al. *Federated Learning with Personalization Layers*. arXiv preprint arXiv:1912.00818, 2019.

[5] Houlsby, N., et al. *Parameter-Efficient Transfer Learning for NLP*. Proceedings of the 36th International Conference on Machine Learning (ICML), 2019.

[6] Oord, A. v. d., et al. *Representation Learning with Contrastive Predictive Coding*. arXiv preprint arXiv:1807.03748, 2018.

[7] McAuley, J., et al. *Ups and Downs: Modeling the Visual Evolution of Fashion Trends with One-Class Collaborative Filtering*. Proceedings of the 25th International World Wide Web Conference (WWW), 2016.

[8] Kingma, D. P., and Ba, J. *Adam: A Method for Stochastic Optimization*. International Conference on Learning Representations (ICLR), 2015.

[9] European Parliament. *General Data Protection Regulation (GDPR)*. Official Journal of the European Union, 2016.